

ILDAR SHAYAKHMETOV

Senior Backend & AI-Systems Engineer

+381 62 975 4956 il.sach@yandex.ru ildarrash.github.io/IldarRash Belgrade, Serbia

SUMMARY

Senior Backend & AI-Systems Engineer with 6+ years designing high-load, distributed backend systems. Expert in Java, Kotlin, and Scala with reactive programming (Project Reactor, Akka Streams) and deep fintech experience — payments, money movement, and PCI-DSS-compliant services. Also architects event-driven AI platforms in Rust and Python (algorithmic trading, AI health coaching). Track record of leading projects, optimizing performance, and keeping systems stable under load

KEY ACHIEVEMENTS

Engineered a real-time streaming platform processing 5M+ messages/sec from Kafka (Java + Akka Streams)

Describe what you did and the impact it had.

Reduced memory footprint and CPU time by ****80%**** via a GraalVM native-image migration, cutting cloud spend by ****~EUR 20k/month****.

Describe what you did and the impact it had.

Delivered PCI-DSS-compliant payment services in production (Kotlin, Project Reactor)

Describe what you did and the impact it had.

Architected an event-driven algorithmic-trading platform with an architecturally unbypassable risk engine (Rust + Python + Kafka)

Describe what you did and the impact it had.

EXPERIENCE

Software Developer

Paysend

02/2023 - Present Belgrade, Serbia

- Develop and maintain high-load payment services in Kotlin and Java; reactive architecture with Project Reactor
- Ensure PCI-DSS compliance and transaction security; integrate external payment systems
- Manage inter-service messaging over RabbitMQ; drove a GraalVM native-image migration saving ~EUR 20k/month
- Build CI/CD pipelines (GitLab) with automated testing (JUnit, TestContainers)

Software Engineer

Perfect Art

05/2022 - 02/2023 Belgrade, Serbia

- Architected a financial-activity screening system for small businesses
- Implemented core logic in Scala + Akka for a highly concurrent, resilient solution

Software Engineer

MTS

05/2020 - 05/2022 Moscow, Russia

- Designed a real-time, high-load streaming platform processing 15M+ messages/sec from Kafka
- Built the core processing engine in Java + Akka Streams; wrote consumer services in Scala (ScalaTest)
- Used Aerospike for caching and deduplication; managed inter-service communication through Kafka

Software Engineer

Raiffeisen Bank

05/2018 - 08/2019 Moscow, Russia

- Built and enhanced a high-load notification service in Java + Spring Integration; rewrote the mass-mailing flow
- Redesigned caching mechanisms to improve performance; built a notification-history service from scratch
- Contributed to frontend work with React and WebSockets for real-time communication

Earlier roles

2016 - 2019

- Java & Android roles at Alfa-Bank, EPAM Systems, Trendsoft, and Dixy — core banking (OpenID Connect authentication), high-load data-validation systems and REST APIs, system integrations (1C, AmoCRM), and mobile development (Retrofit, RxJava)

EDUCATION

B.Sc., Software Engineering

HSE University

2014 - 2018 Location

LANGUAGES

English

Advanced (C1)



Russian

Native



SKILLS

Spring Boot	Spring Integration	Project Reactor	Akka Streams	ZIO	Tokio	RSocket	NestJS
Messaging & Streaming							
Apache Kafka	RabbitMQ						
Data & Caching							
PostgreSQL	Aerospike	Redis					
AI & Event-Driven							
event-driven architecture	gRPC	LLM-backed systems	auditable/reproducible decision pipelines				
DevOps & Tooling							
Docker	GitLab CI/CD	Gradle	TestContainers	JUnit	Git	GraalVM	

PROJECTS

HealthTracer

AI health coach built as a system, not a chatbot: a Postgres-backed structured domain model is authoritative, while the AI emits **typed proposals** (never direct DB writes) that the backend validates and the user approves, each becoming an immutable, auditable revision. The core is a multi-domain fan-out & synthesis LLM pipeline (a router selects up to 3 domains, runs them in parallel, then a decision-maker synthesizes their output into proposals) with a deterministic planner clamping output to a hard capability allowlist and safety floors enforced in code. **TypeScript, NestJS, Next.js, Expo, Drizzle/PostgreSQL, Turborepo.**

BeoSand

Telegram-first platform for booking volleyball courts and managing training in Belgrade: a React admin console (scheduling, court-load heatmap, booking requests, broadcasts) plus a Telegram mini-app for players (browse slots, book, waitlist). Server-authoritative by design — all pricing, availability, and capacity are computed server-side and the UI only displays values and collects actions. Telegram Login auth, RSD pricing, RU/SR/EN i18n.

CERTIFICATIONS

JVA-074 Java Advanced: Functional, Asynchronous and Reactive Programming	Kotlin Developer
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FIND ME ONLINE



GitHub
github.com/IldarRash